

SOFIANE MALLEM

Python Software Developer

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LinkedIn Github Portfolio

SUMMARY

Python developer experienced in building and operating production systems: scaled rendering infrastructure, automation tooling, and data pipelines feeding ML training. Owned a CAD-to-rendering workflow end to end, from customer data ingestion to downstream model training. Comfortable working with AI coding tools (Claude Code, Cursor) in daily development.

EXPERIENCE

Synthavo

Nov 2025 – Apr 2026

Python Developer

Stuttgart, BW

- Owned the customer CAD data pipeline, validating, transforming, and uploading incoming datasets powering a horizontally scaled Blender rendering service across 5+ Linux servers.
- Migrated the rendering service to a newer Blender version, improving render speed by ~40% on top of prior optimization work, while sustaining over 99% uptime through proactive monitoring and bottleneck resolution.
- Maintained the rendering-to-training pipeline feeding YOLOv5 training and coordinated delivery of trained models to QA team.
- Operated the deployment layer with Ansible (pushing container images, adjusting instances, assigning resources) within an existing Infrastructure-as-Code setup.
- Primary technical contact for production incidents and customer requirements, diagnosing failures and scoping tasks.

Synthavo

Feb 2025 – Aug 2025

Python Developer Intern

Stuttgart, BW

- Developed Python tooling for the CAD validation and transformation pipeline, improving rendering reliability.
- Applied mesh reduction to incoming STL files ahead of rendering, improving render throughput by approximately 40%.
- Cut customer onboarding provisioning time from ~5 days to 2–3 days through Python tooling and process automation.

PROJECTS

LAD (Local AI Daemon) | Self-Hosted AI Assistant with Tool-Calling Agent

GitHub

Stack: FastAPI · React/TypeScript · PostgreSQL + pgvector · Ollama · MCP · SSE · Docker Compose · Nginx · GitHub Actions

- Built a fully local ChatGPT/Claude-style assistant (streaming replies, persistent chats, file uploads) on Ollama models with no external model API, run with Docker Compose behind Nginx.
- Designed a tool-calling agent loop on the Model Context Protocol, each tool (web search, memory recall, shell) running as its own MCP server; chat and agent modes use separate models, prompts, and tool allowlists enforced server-side.
- Built an isolated command-execution sandbox service (separate container, non-root user, CPU/memory caps, per-chat working directories), verified free of cross-chat contamination under concurrent use.
- Implemented cross-chat semantic memory using embedded rolling summaries and pgvector similarity search; covered by unit and integration tests (real Postgres via Testcontainers) and three CI workflows.

Hang | Scalable Distributed Messaging Platform

GitHub

Stack: FastAPI · WebSockets · Redis Pub/Sub · PostgreSQL · Docker · Nginx · OAuth · S3 · Prometheus · React/TS

- Developed a real-time messaging platform with authentication, persistent messages, typing indicators, read receipts, media uploads, and cursor-based pagination for large histories.
- Architected a horizontally scalable WebSocket layer using Redis Pub/Sub to synchronize events across stateless instances behind an Nginx load balancer; integrated Google OAuth and S3-compatible storage, and load-tested a production-like local environment (Docker, PostgreSQL, Redis, Prometheus).
- Directed AI-assisted development (Claude Code) for the React frontend; owned architecture, data flow, and backend.

EDUCATION

42 Heilbronn

Feb 2023 – Nov 2025

Software Engineering Course

Heilbronn, Germany

University of S.T Houari Boumediene

2017 – 2020

Bachelor of Computer Science

Algiers, Algeria

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, TypeScript, C, C++, Bash

Backend: FastAPI, REST APIs, WebSockets, OAuth, Async Python

ML & AI: YOLOv5 training pipelines, Object Detection, AI-Assisted Development (Claude Code, Cursor)

Data: PostgreSQL, Redis, Pandas, Data Pipelines, CAD Processing

Infrastructure: Docker, Linux, Nginx, CI/CD, GitHub Actions, Google Cloud Platform (Cloud Run), Ansible, Prometheus

Practices: Git, Pytest, Unit/Integration Testing, Agile

LANGUAGES

English (Fluent) | French (Native) | Arabic (Native) | German (A1)